Swiss Institute of Bioinformatics

PR STATS

Metagenomic Data Analysis of Microbial Communities (MGMC01)

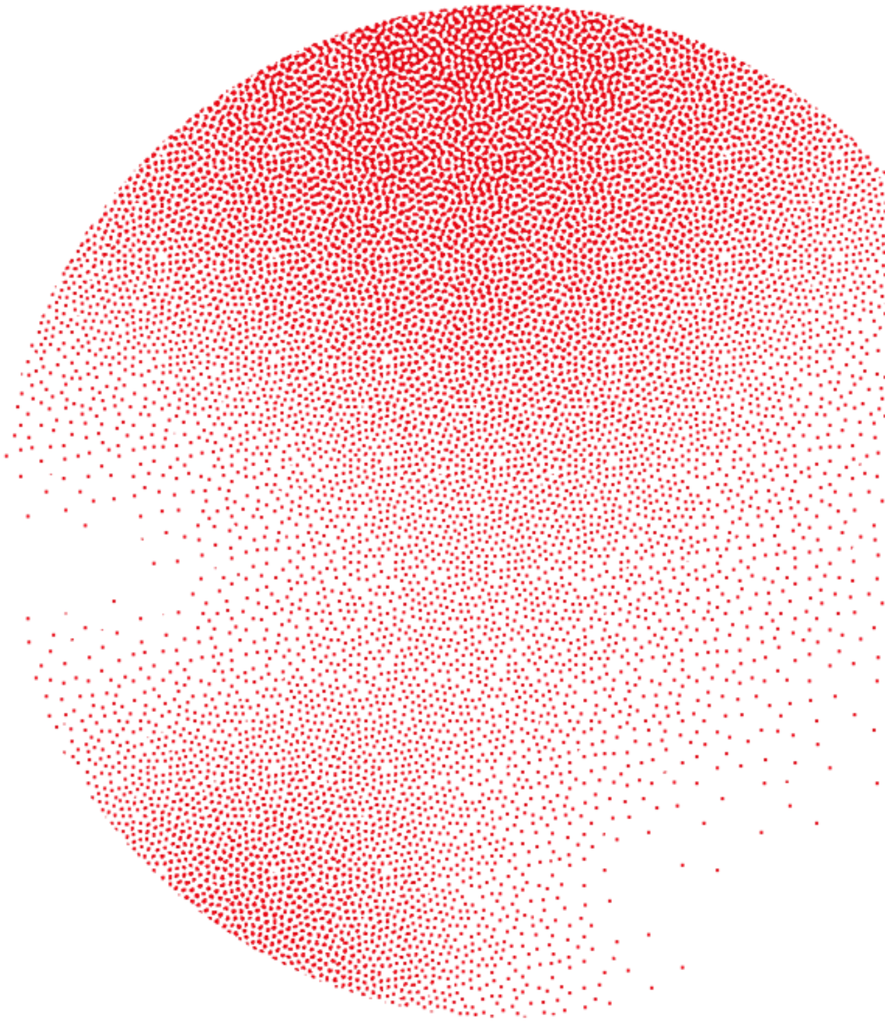
Day 5 - Session 1

PhD Jeferyd Yepes-García
University of Fribourg
October 30th, 2026

elixir SWITZERLAND

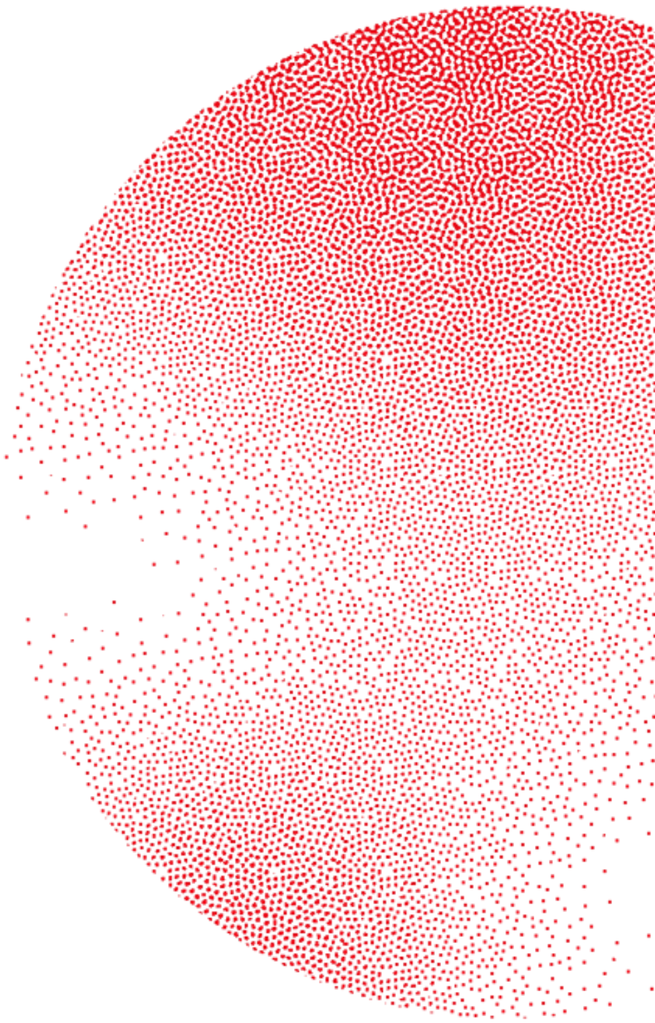
This presentation

- Learning outcomes
- Reproducibility crisis
- Methodologies for reproducible research
- Workflow managers and containers
- TaxoFlow as a case of study
- Open Access and Data Management
- Submission to NCBI
- Summary



Learning outcomes

- Identify different methodologies to enable reproducible research.
- Understand the requirements that every project must fulfill in order to be FAIR.
- Recognize the benefits of making projects Open Access.
- Describe the main steps to follow in order to submit both raw and curated data the NCBI repositories.



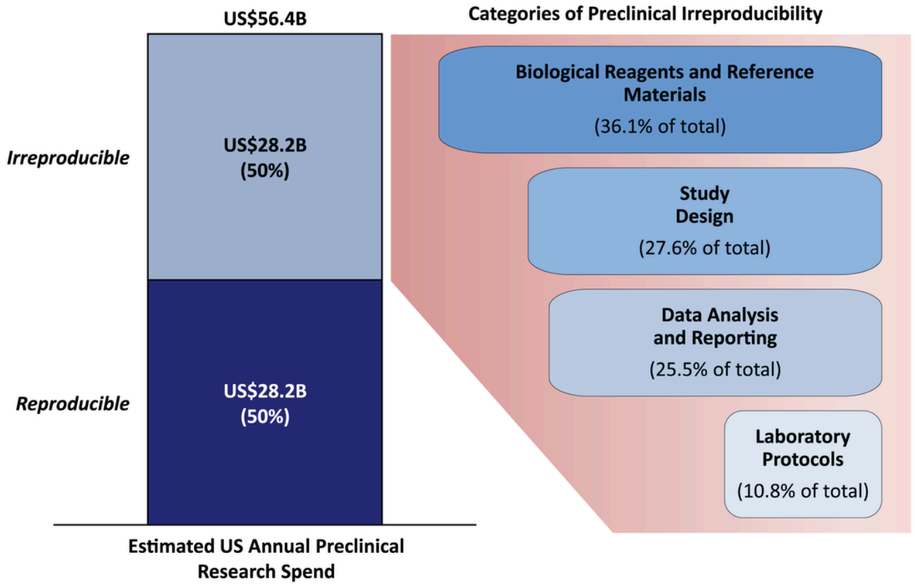
Reproducibility crisis?

PERSPECTIVE

The Economics of Reproducibility in Preclinical Research

Leonard P. Freedman[1]*, Iain M. Cockburn[2], Timothy S. Simcoe[2,3]

1 Global Biological Standards Institute, Washington, D.C., United States of America, **2** Boston University School of Management, Boston, Massachusetts, United States of America, **3** Council of Economic Advisers, Washington, D.C., United States of America



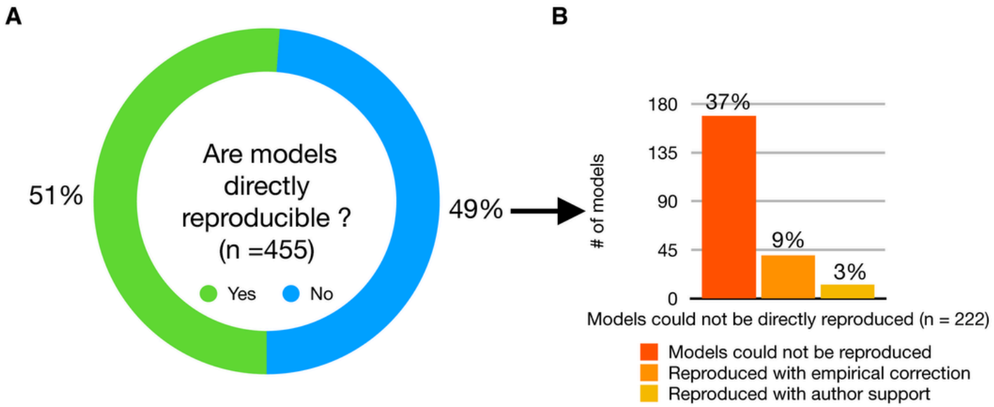
Commentary

OPEN ACCESS

molecular systems biology

Reproducibility in systems biology modelling

Krishna Tiwari[1,2], Sarubini Kananathan[1], Matthew G Roberts[1], Johannes P Meyer[1], Mohammad Umer Sharif Shohan[1], Ashley Xavier[1], Matthieu Maire[1], Ahmad Zyoud[1], Jinghao Men[1], Szeyi Ng[1], Tung V N Nguyen[1], Mihai Glont[1], Henning Hermjakob[1,3,*] & Rahuman S Malik-Sheriff[1,**]



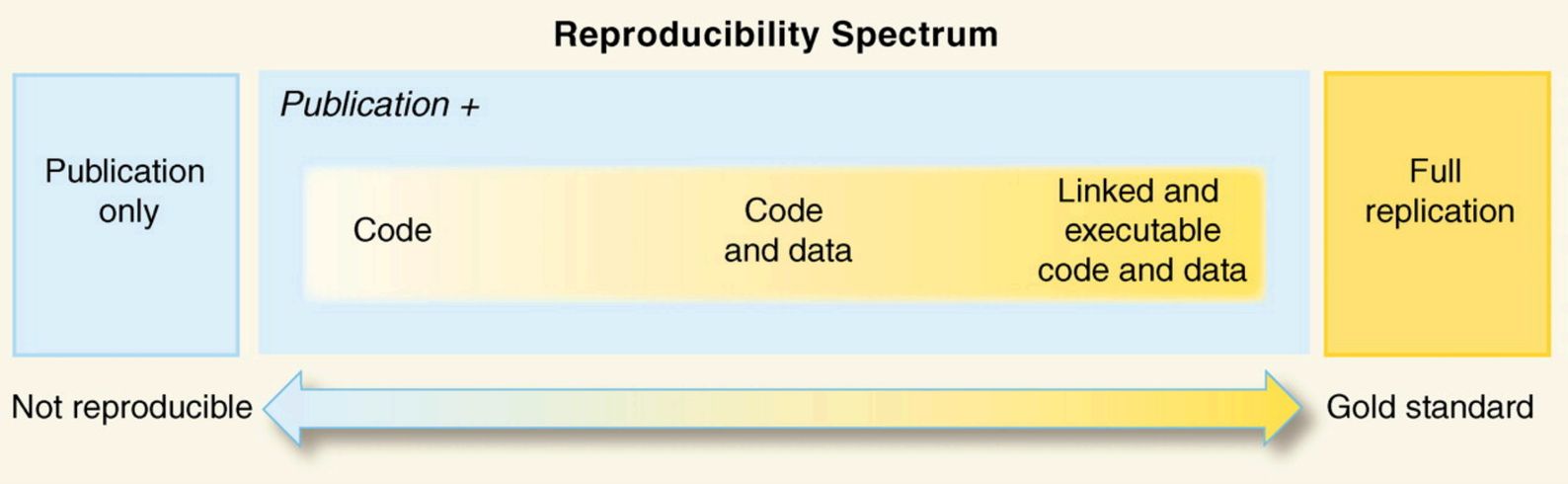
Reproducibility in computational biology and bioinformatics

TABLE 1 Simple grid-based system for defining concepts that can be used to describe the validity of a result[a]

Methods	Same experimental system	Different experimental system
Same methods	Reproducibility	Replicability
Different methods	Robustness	Generalizability

[a]This is a generalization of the approach used by Whitaker (9), who used it to describe computational analyses.

Schloss (2018)



Peng (2011)

Reproducibility in computational biology and bioinformatics

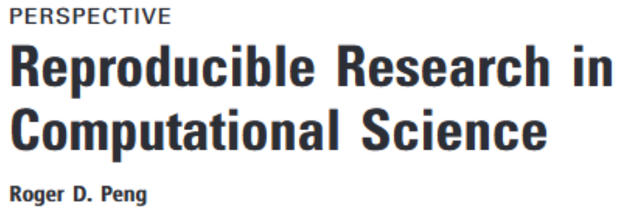
In computational biology and bioinformatics, reproducibility implicitly means the ability to obtain **identical outputs** from the **same code, data, and software environment** (Ziemann et al. 2023).

```
Output = Algorithm (Code) + Data + Parameters
```

Problems

- Versioning.
- Data availability.
- Environment.
- Parameters setting.
- Hidden manipulation steps.
- Random seeds.
- ...

Strategies to enable reproducibility in computational research



PERSPECTIVE

Reproducible Research in Computational Science

Roger D. Peng



PERSPECTIVE

Good enough practices in scientific computing

Greg Wilson[1]*, Jennifer Bryan[2], Karen Cranston[3], Justin Kitzes[4], Lex Nederbragt[5], Tracy K. Teal[6]



OPEN ACCESS Freely available online

PLOS COMPUTATIONAL BIOLOGY

Editorial

Ten Simple Rules for Reproducible Computational Research



Check for updates

OPEN ARTICLE

Introducing the FAIR Principles for research software

The FAIR Principles

Findable



To be Findable:
F1. (meta)data are assigned a **globally unique and persistent identifier**
F2. data are described with **rich metadata** (defined by R1 below)
F3. metadata **clearly and explicitly include the identifier of the data it describes**
F4. (meta)data are registered or **indexed in a searchable resource**

Accessible



To be Accessible:
A1. (meta)data are retrievable by their identifier using **a standardized communications protocol**
A1.1 the protocol is **open, free, and universally implementable**
A1.2 the protocol allows for an **authentication and authorization procedure**, where necessary
A2. **metadata are accessible**, even when the data are no longer available

Interoperable



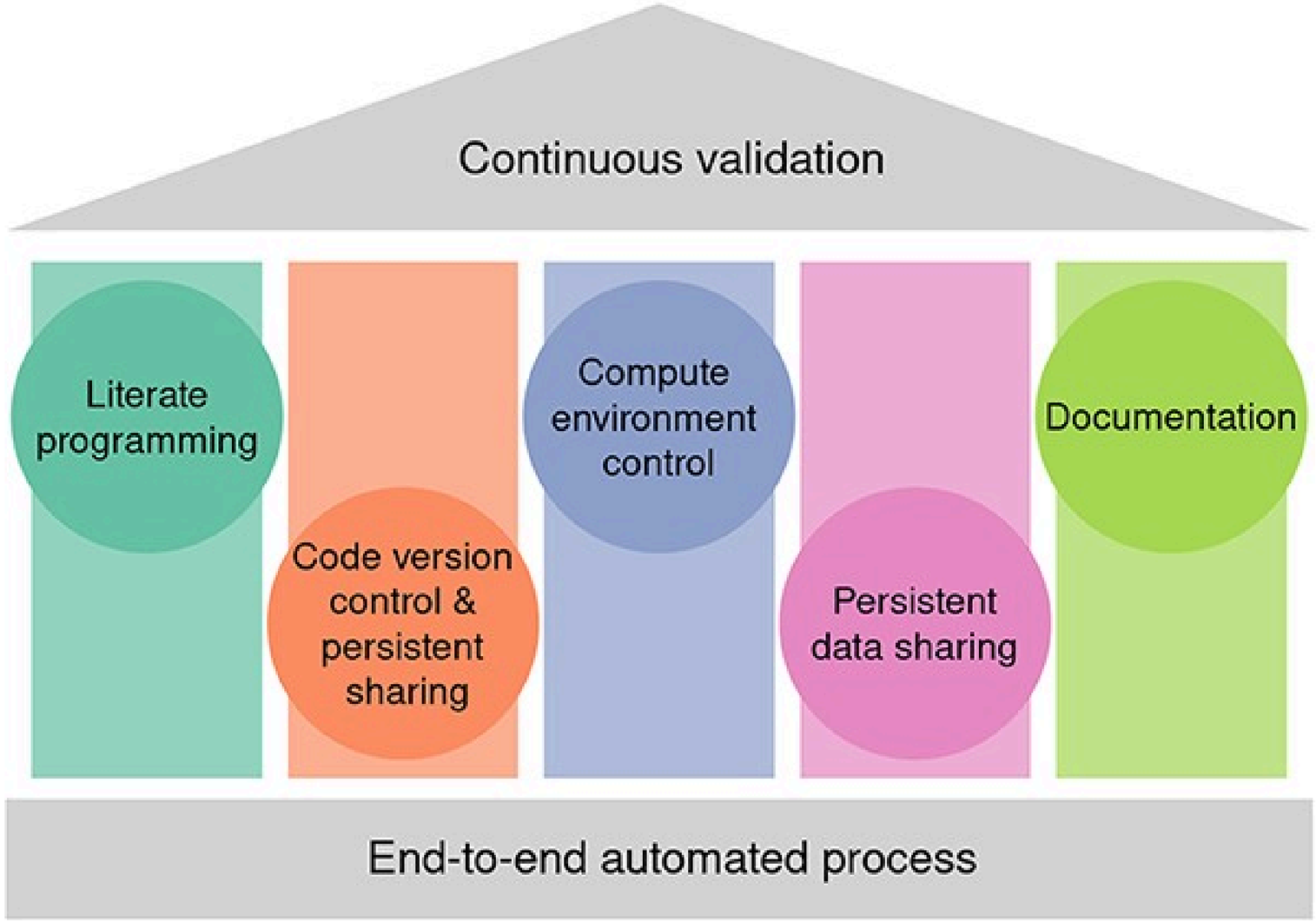
To be Interoperable:
I1. (meta)data use a **formal, accessible, shared, and broadly applicable language for knowledge representation.**
I2. (meta)data use **vocabularies that follow FAIR principles**
I3. (meta)data include **qualified references to other (meta)data**

Reusable



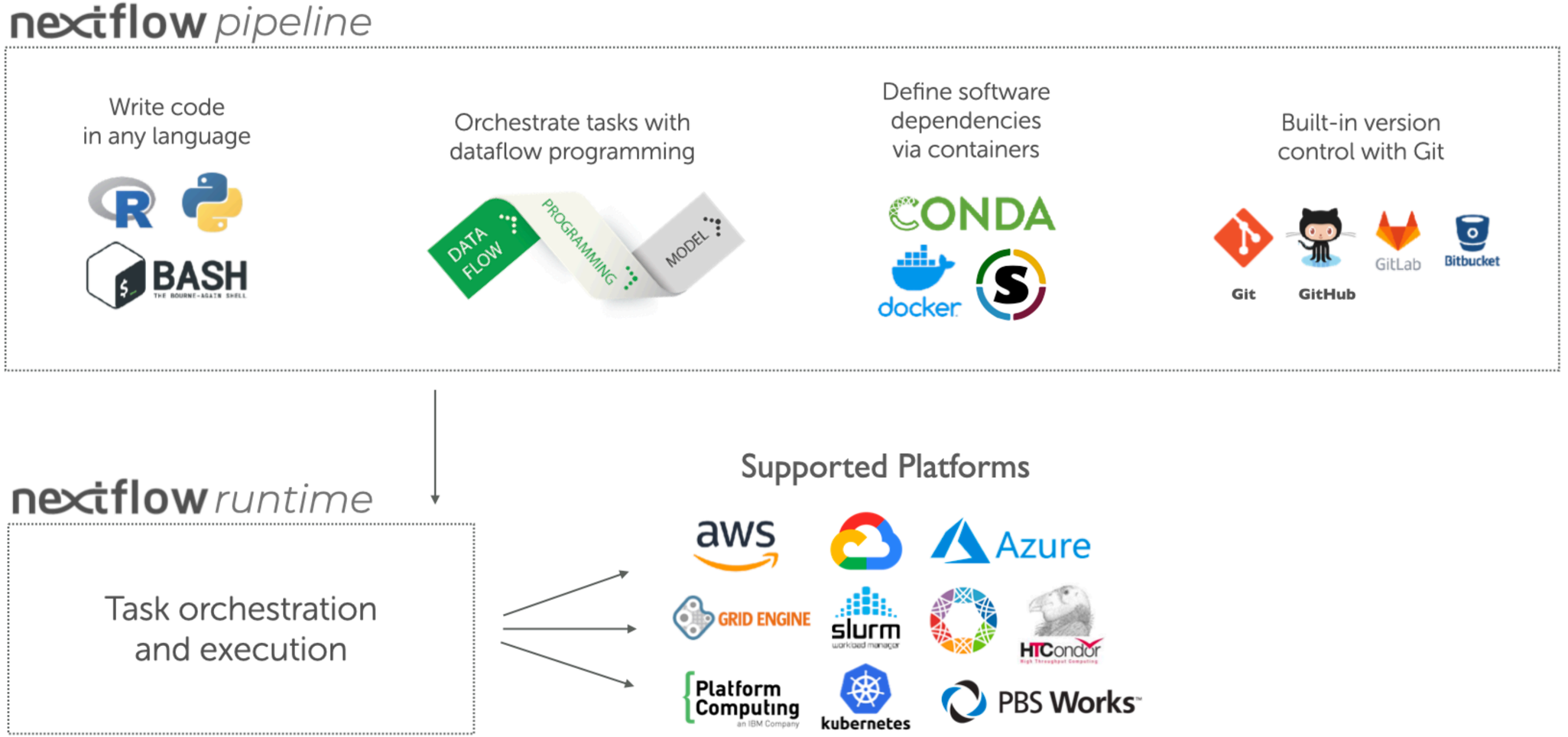
To be Reusable:
R1. meta(data) are **richly described with a plurality of accurate and relevant attributes**
R1.1. (meta)data are released with a **clear and accessible data usage license**
R1.2. (meta)data are associated with **detailed provenance**
R1.3. (meta)data meet domain-relevant **community standards**

Five pillar of reproducible computational research



Ziemann et al. (2023)

Workflow managers and containers/package managers



TaxoFlow as case of study

Findable

- Persistent identifier (DOI)
- Available ORCIDs
- GitHub-linked
- Deposited on Zenodo

Accessible

- Open Access
- Standard protocol: (HTTPS, git, Zenodo)
- Implementation in different platforms (CodeSpaces, CodeSandbox)



Interoperable

- Community-standard formats (FASTQ, BIOM)
- Workflow manager (Nextflow)
- SRA accessions

Reusable

- Documentation (paper)
- Pinned versions (containers)
- Validation (mock community)
- Supported reference

TaxoFlow as case of study

Findable

- Registration in specialized repositories (bio.tools)
- Different identifiers

Accessible

- Codespaces and CodeSandbox are commercial platforms



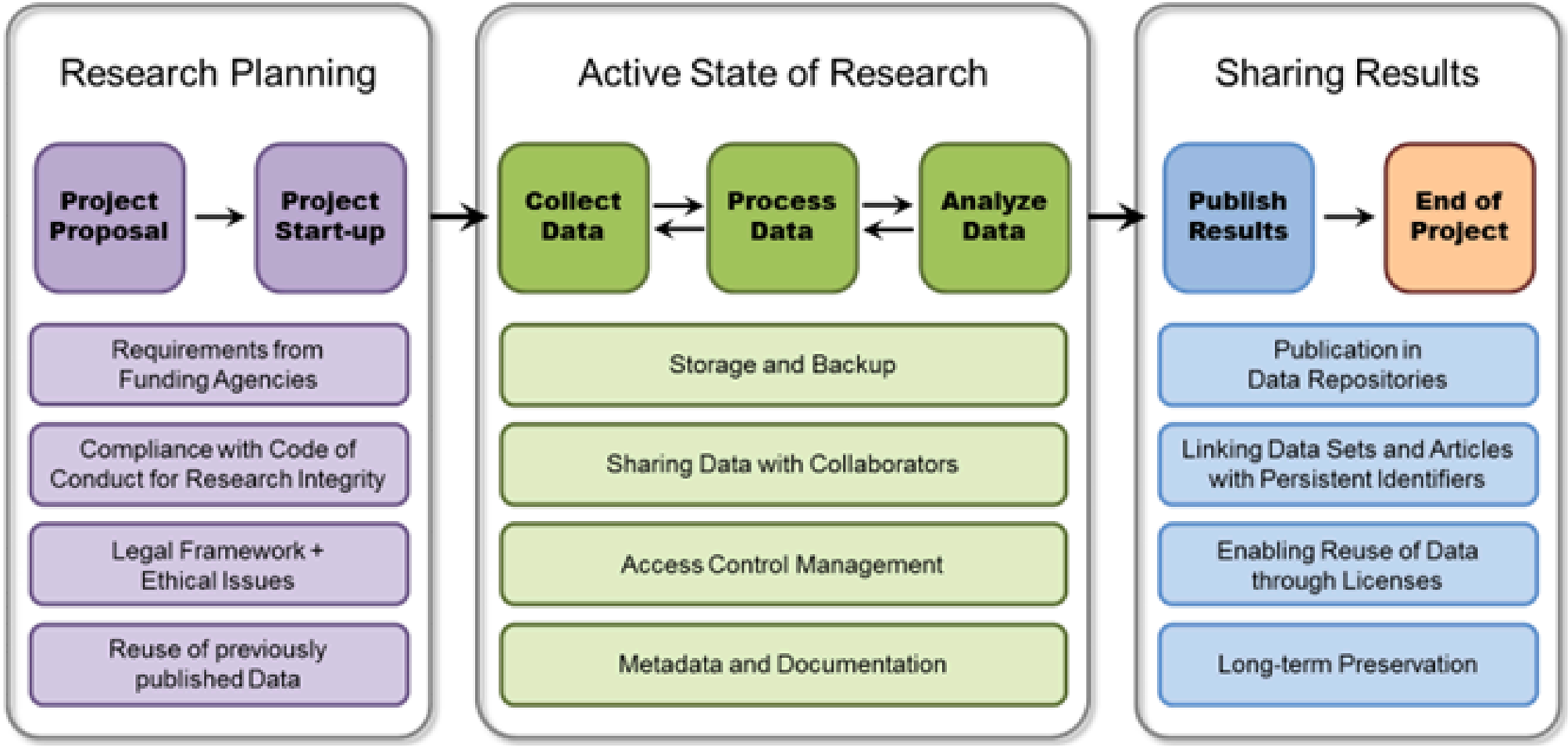
Interoperable

- Not fully compliant of nf-core standards
- Schema-validated samplesheet

Reusable

- License difference (tool and paper)
- Automated CI

Data Management Plan (DMP)



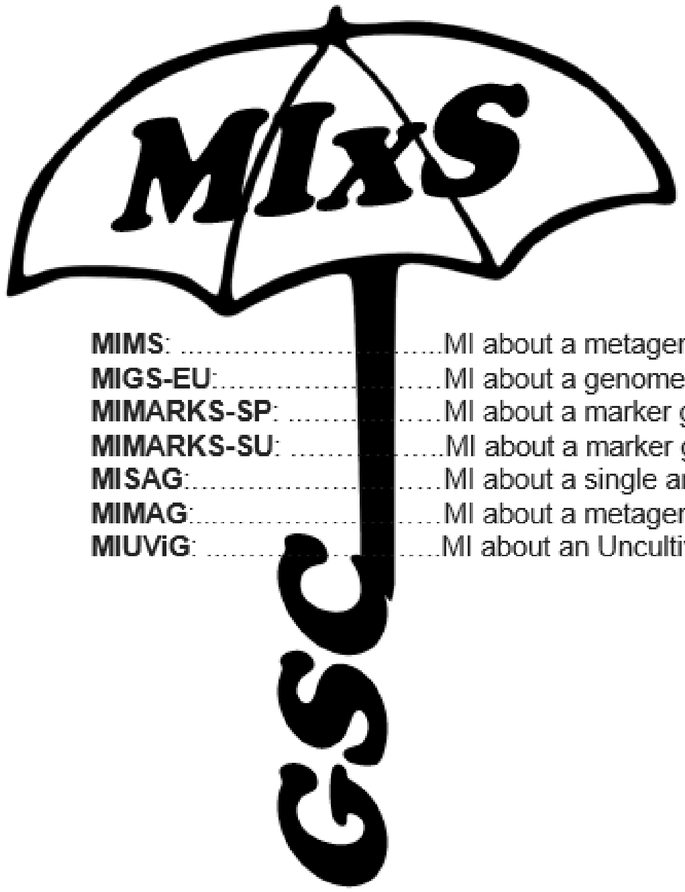
Benefits to Open Research Data

Direct Verification of published research Preserving accessibility to data Allowing reuse and repurposing of data Discoverability of data	**Indirect (cost avoid)** Redundant data collection Inefficient legacy data curation Burden of sharing-upon-request Studies cannot be completed
Short-term Availability for review Availability for validation	**Long-term** Persistent link with article data Increased impact per publication
To the author Protection against data entropy Improved methodologies Higher diffusion and visibility Higher citation rate of their publications Fulfillment of funding mandate	**To the scientific community & public** More efficient use of research fundings Foster collaboration Accelerate innovation Educational opportunities Public trust in science

MIxS - MIMAG

Minimum Information about any (x) Sequence

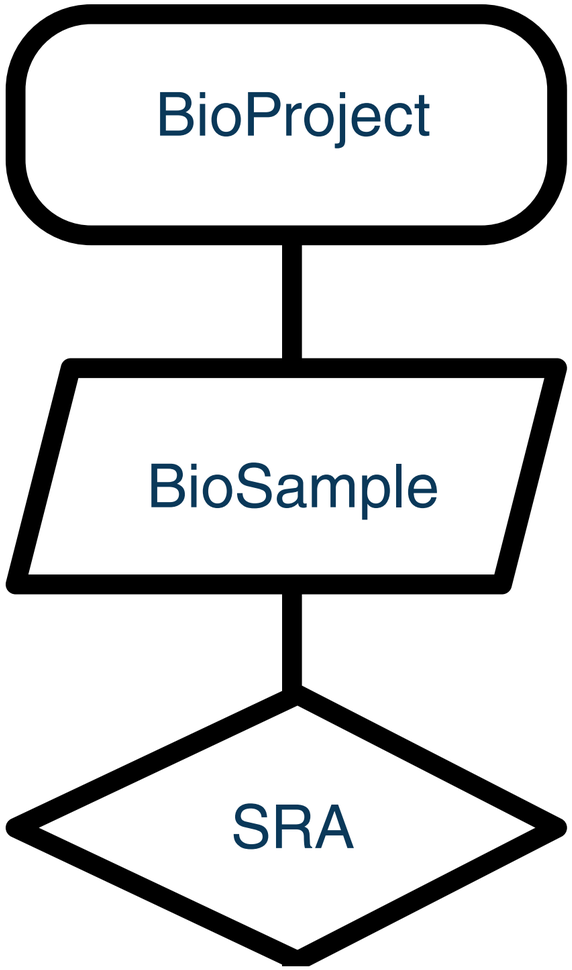
Minimum Information About a Metagenome-Assembled Genome



MIMS: .....MI about a metagenome sequence.
MIGS-EU:.....MI about a genome sequence
MIMARKS-SP: .....MI about a marker gene sequence obtained directly from the environment
MIMARKS-SU: .....MI about a marker gene sequence from cultured or voucher-identifiable specimens
MISAG:.....MI about a single amplified genome sequence.
MIMAG:.....MI about a metagenome-assembled genome sequence.
MIUViG: .....MI about an Uncultivated Virus Genome

Submission to the Sequence Read Archive

Raw reads:



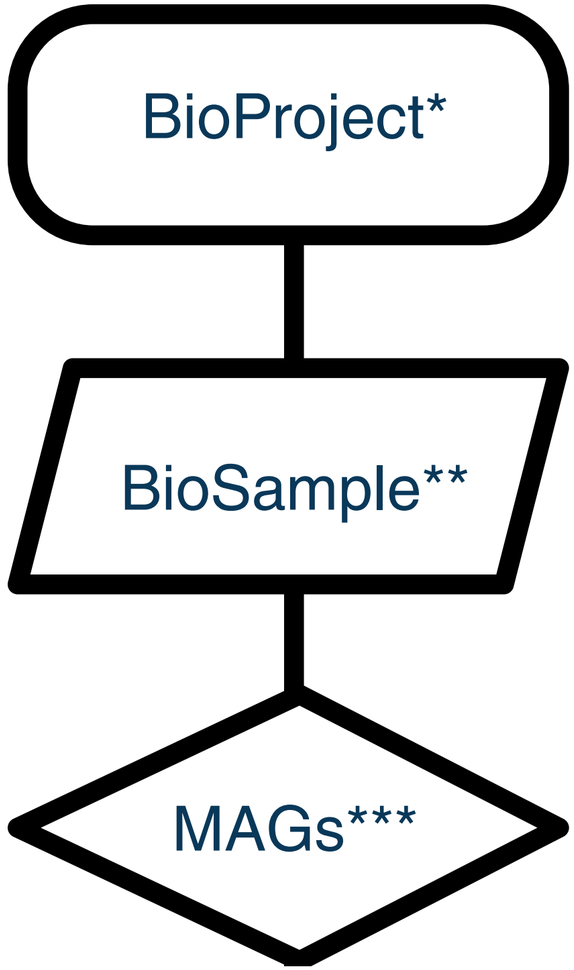
Goals, participants, grants, publications, etc.

Type of material, location, etc.

Sequencing platform, library, layout, etc.

Submission to the NCBI Genome

High-quality MAGs:



Goals, participants, grants, publications, etc.

Organism name, location, Environmental Ontology, etc.

Assembly method, coverage, etc.

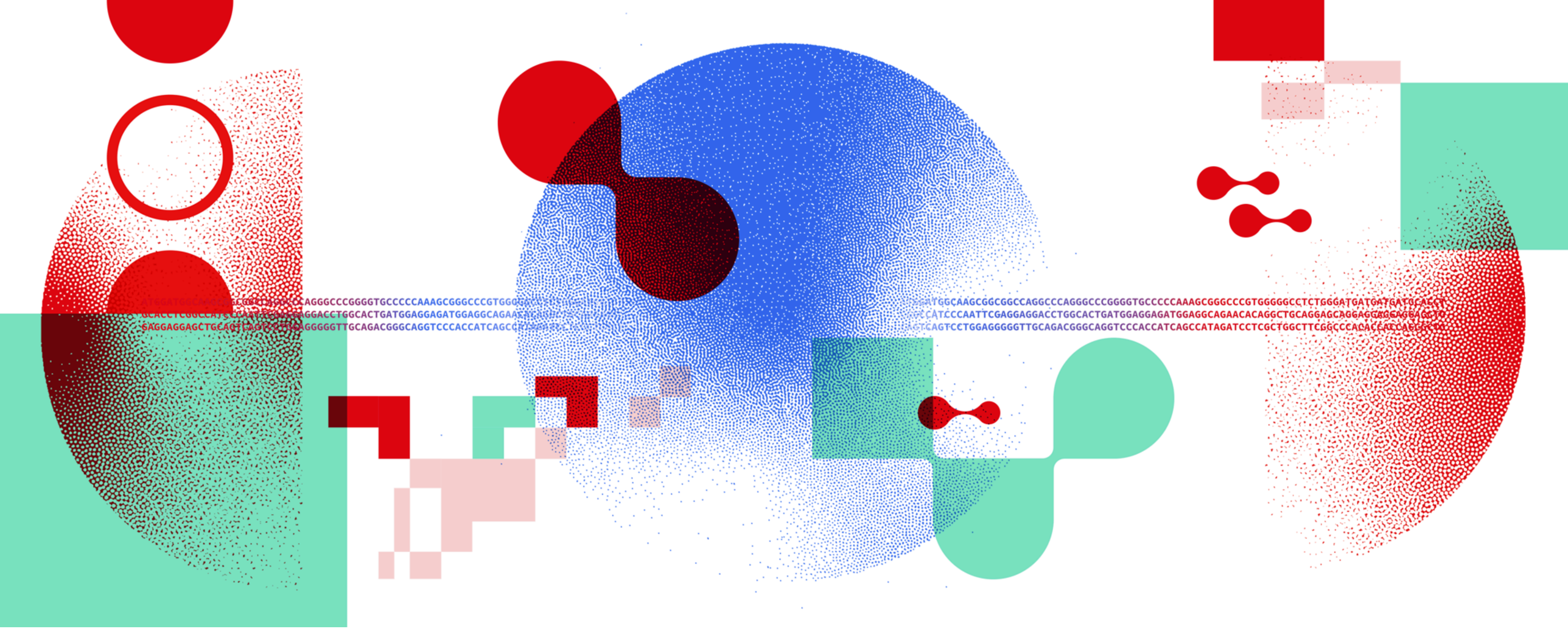
***Same as raw reads.**
****Different from raw reads.**
*****Names, quality metrics and SRA provenance.**

Resources and tutorials:

- BioProject and BioSamples for raw reads.
- Sequence Read Archive (SRA) submission.
- Metagenome submission guide.
- MAG submission.

Summary

- Reproducibility crisis and science is real and must be addressed.
- There exist several approaches to support reproducibility and study reliability such as the FAIR principles and the five pillars of reproducible computational research.
- Open Access projects provide benefits to both the community and the particular researchers.



Thank you

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SIB